

		Direct seq filter	Closest serialize injective	Unstable same-view serialize	Hard range consistent	Aggregate ETtS Δ vs H0 (%) ← fewer tokens			
						0	25	50	75
	H0 no skill	25.6k baseline	71.8k baseline	32.8k baseline	79.2k baseline	52.4k	4/4		
R1	A bounded exploration gate	28.1k +10.1%	78.7k +9.6%	29.1k -11.1%	70.0k -11.7%	51.5k	4/4		
	R1 C delta certificate	28.4k +11.3%	66.0k -8.1%	50.1k +52.9%	70.6k -11.0%	53.8k	4/4		
	R1 S obligation graph	28.6k +11.9%	72.7k +1.3%	20.0k -39.0%	88.4k +11.5%	52.4k	4/4		
R2	A diagnostic transaction	30.6k +19.8%	NO USAGE	36.0k +9.8%	NO USAGE	×	invalid	2/4	
R2	C contract first fast path	40.3k +57.6%	NO USAGE	55.6k +69.7%	91.4k +15.3%	×	invalid	3/4	
R2	S branch witness cutset	31.9k +24.8%	84.6k +17.9%	19.9k -39.3%	98.9k +24.8%	58.8k	4/4		
R3	A failure budget pivot	59.1k +131.3%	118.1k +64.4%	36.0k +9.8%	96.8k +22.1%	77.5k	4/4		
R3	C local contract closure	32.3k +26.5%	73.7k +2.6%	18.5k -43.5%	89.1k +12.4%	53.4k	4/4		
R3	S inline branch certificates	19.8k -22.6%	NO USAGE	38.8k +18.3%	62.2k -21.5%	×	invalid	3/4	
R4	A single live hypothesis	34.2k +33.7%	FAIL 82.6k	56.5k +72.2%	106.7k +34.7%	93.3k	3/4		
R4	C zero ceremony direct	19.6k -23.2%	72.9k +1.5%	28.1k -14.2%	115.5k +45.7%	59.0k	4/4		
R4	S return polarity slicing	21.0k -18.0%	69.8k -2.8%	56.5k +72.5%	121.8k +53.7%	67.3k	4/4		
R5	A batched compiler oracle	30.7k +20.0%	FAIL 77.2k	30.1k -8.2%	56.0k -29.3%	64.7k	3/4		
	R5 C direct self disable	43.9k +72.0%	73.8k +2.8%	31.3k -4.5%	74.7k -5.8%	55.9k	4/4		
R5	S backward contract frontier	29.6k +15.7%	64.2k -10.6%	35.5k +8.4%	78.8k -0.6%	52.0k	4/4		
R6	A branch certificate cutoff	24.9k -2.5%	NO USAGE	29.0k -11.6%	71.7k -9.5%	×	invalid	3/4	
	R6 C micro direct kernel	31.1k +21.9%	66.9k -6.9%	33.0k +0.7%	76.5k -3.4%	51.9k	4/4		
R6	S typed two stage oracle	30.6k +19.6%	58.5k -18.6%	30.2k -7.9%	94.8k +19.6%	53.5k	4/4		